

# Finite-circle curvature and arithmetic control

Split-Zero research programme

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This companion contains the complete exponential-period determinant proof, the constituent-curvature calculation, the original Laplacian control, and the finite-source comparison through the exact metric commutator. Every original equation label, coefficient frame, source metric and operator order is retained.

The editable source package includes the complete supporting periodization, density, residue and original-source proofs, together with their source inventory and review evidence. The circle realization retains its actual remainder map, Fourier zero coordinate and theta boundary; the arithmetic quantities in the estimates remain the ones defined in those original objects.

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# 1 Exact determinant of the polynomial–exponential period comparison

Fix the original monic cyclic polynomial

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad q \geq 1, \quad d = q + 1, \quad \Phi_t(S) = \frac{S^d}{d} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS.$$

Fix  $u \in \mathbb{C}^*$  and one argument of  $u$ . All the following comparisons retain this  $u$ , the original coordinate  $S$ , and the ordered remainder basis  $1, S, \dots, S^{q-1}$ . Set

$$D_{u,t}P = uP' + (\chi - t)P, \quad H_{u,t} = \mathbb{C}[S]/D_{u,t}\mathbb{C}[S].$$

This is a quotient of coefficient vector spaces. Let  $\text{red}_{u,t}$  be its unique representative of degree less than  $q$ . Let  $A(t)$  be the multiplication matrix of  $S$  on  $\mathbb{C}[S]/(\chi - t)$  in the same ordered basis, so

$$A(t)e_b = e_{b+1} \quad (0 \leq b < q-1), \quad A(t)e_{q-1} = (-c_0 + t)e_0 - \sum_{a=1}^{q-1} c_a e_a. \quad (\text{XD1})$$

The empty sum convention in this display retains the case  $q = 1$ .

Use precisely the oriented rays and differences

$$\theta_j = \frac{\arg u + \pi}{d} + \frac{2\pi j}{d}, \quad \ell_j : S = re^{i\theta_j}, \quad 0 \leq r < \infty, \quad \Gamma_j = -\ell_0 + \ell_j \quad (1 \leq j \leq q). \quad (\text{XD2})$$

Write

$$\Pi_{jb}(u, c, t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS, \quad 1 \leq j \leq q, \quad 0 \leq b < q. \quad (\text{XD3})$$

## 1.1 Uniform estimates on the fixed unbounded contours

For a compact set  $K \subset \mathbb{C}^{q+1}$  of the parameters  $(c_0, \dots, c_{q-1}, t)$ , put

$$M_K = \sum_{a=0}^{q-1} \frac{\sup_K |c_a|}{a+1} + \sup_K |t|, \quad R_K = \max\{1, 2dM_K\}, \quad \alpha = \frac{1}{2d|u|}. \quad (\text{XD4})$$

For every ray in (XD2) its leading term satisfies  $S^d/(du) = -r^d/(d|u|)$ . For  $r \geq 1$  every other exponent has degree at most  $d-1$ , so

$$\text{Re} \frac{\Phi_t(re^{i\theta_j})}{u} \leq -\frac{r^d}{d|u|} + \frac{M_K r^{d-1}}{|u|} \leq -\alpha r^d \quad (r \geq R_K). \quad (\text{XD5})$$

Every parameter derivative of the integrand is its original exponential times a polynomial in  $S$ : a derivative in  $c_a$  contributes  $S^{a+1}/((a+1)u)$  and a derivative in  $t$  contributes  $-S/u$ . For any fixed derivative order its absolute value on the tail is bounded by  $C r^L e^{-\alpha r^d}$ , with  $C, L$  uniform on  $K$ . The required majorant is integrable, since substitution  $x = \alpha r^d$  gives

$$\int_0^\infty r^L e^{-\alpha r^d} dr = \frac{1}{d} \alpha^{-(L+1)/d} \Gamma\left(\frac{L+1}{d}\right) < \infty. \quad (\text{XD6})$$

The remaining finite interval is uniformly bounded by continuity. Thus (XD3) converges absolutely and is entire in all the lower coefficients and  $t$ ; every parameter derivative may be taken under the integral. The same bound gives  $e^{\Phi_t(S)/u} P(S) \rightarrow 0$  at the infinite endpoint of each ray, for every polynomial  $P$ . Consequently

$$\int_{\Gamma_j} e^{\Phi_t(S)/u} D_{u,t}P dS = u \int_{\Gamma_j} d(e^{\Phi_t(S)/u} P) = 0. \quad (\text{XD7})$$

Indeed the integral of that derivative on each outward ray is  $-e^{\Phi_t(0)/u} P(0) = -P(0)$ ; the coefficients  $-1, +1$  of the two rays cancel these values. These estimates handle the noncompact contours directly, including parameter values at which critical points coincide.

## 1.2 An exact trace comparison for twisted reduction

For  $1 \leq m \leq q$  and  $0 \leq b < q$ , ordinary monic division gives unique polynomials  $Q_{m,b}, R_{m,b}$  satisfying

$$S^{m+b} = (\chi - t)Q_{m,b} + R_{m,b}, \quad \deg R_{m,b} < q. \quad (\text{XD8})$$

If  $m + b < q$ , take  $Q_{m,b} = 0$ . Otherwise  $\deg Q_{m,b} = m + b - q \leq q - 1$ , and the differential itself gives

$$\text{red}_{u,t}(S^{m+b}) = R_{m,b} - uQ'_{m,b}. \quad (\text{XD9})$$

Both terms on the right have degree below  $q$ . In every nonzero case,

$$\deg Q'_{m,b} \leq m + b - q - 1 \leq b - 1. \quad (\text{XD10})$$

Thus the coefficient of  $S^b$  in the derivative correction vanishes. Let  $T_m(u, c, t)$  have column  $b$  equal to the coordinates of the left side of (XD9). The columns  $R_{m,b}$  are exactly those of  $A(t)^m$ . Although (XD9) retains a possibly nonzero correction matrix, (XD10) proves the exact trace identity

$$\text{Tr } T_m(u, c, t) = \text{Tr } A(t)^m \quad (1 \leq m \leq q). \quad (\text{XD11})$$

Set  $B_a = T_{a+1}/((a+1)u)$  for  $0 \leq a < q$ . Differentiation of (XD3), followed by (XD7)–(XD9), now proves

$$\partial_{c_a} \Pi = \Pi B_a, \quad \partial_t \Pi = -\Pi A(t)/u, \quad \text{Tr } B_a = \frac{\text{Tr } A(t)^{a+1}}{(a+1)u}. \quad (\text{XD12})$$

All the coefficient matrices here have polynomial entries in  $c, t$  for this fixed nonzero  $u$ .

## 1.3 The determinant, including every monomial factor

Define the scalar polynomial in the original coefficients

$$\mathcal{F}(c, t) = \text{Tr } \Phi_t(A(t)) = \frac{\text{Tr } A(t)^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a \text{Tr } A(t)^{a+1}}{a+1} - t \text{Tr } A(t). \quad (\text{XD13})$$

There is no choice or enumeration of roots in this definition. For any differentiable matrix  $A$  and integer  $n \geq 1$ , the product rule and cyclic invariance of trace give

$$d \text{Tr } A^n = \sum_{r=0}^{n-1} \text{Tr}(A^r(dA)A^{n-1-r}) = n \text{Tr}(A^{n-1}dA). \quad (\text{XD14})$$

Because  $A(t)$  acts on the actual quotient by  $\chi - t$ , it satisfies  $\chi(A(t)) - tI_q = 0$ . Applying (XD14) to (XD13) therefore gives

$$d\mathcal{F} = \sum_{a=0}^{q-1} \frac{\text{Tr } A(t)^{a+1}}{a+1} dc_a - \text{Tr } A(t) dt + \text{Tr}((\chi(A(t)) - tI_q)dA(t)) = \sum_{a=0}^{q-1} \frac{\text{Tr } A(t)^{a+1}}{a+1} dc_a - \text{Tr } A(t) dt. \quad (\text{XD15})$$

At  $c = t = 0$  the period matrix is

$$\Pi_{j,l-1}^{\text{mon}}(u) = a_l(u)e^{il\theta_0}(\zeta^{jl} - 1), \quad a_l(u) = \frac{(d|u|)^{l/d}}{d}\Gamma(l/d), \quad \zeta = e^{2\pi i/d}, \quad 1 \leq j, l \leq q. \quad (\text{XD16})$$

This follows by substituting  $x = r^d/(d|u|)$  on each oriented ray in (XD3). In particular all  $a_l(u)$  are positive real numbers. For  $V_{jl} = \zeta^{jl} - 1$  one may include  $j = 0$  in the Gram sum because its summand is zero. The complete root-of-unity sums give

$$(V^*V)_{lm} = \sum_{j=0}^{d-1} (\zeta^{-jl} - 1)(\zeta^{jm} - 1) = d\delta_{lm} + d. \quad (\text{XD17})$$

Here  $1 \leq l, m < d$ , so the two individual nonconstant sums vanish, and the first sum is  $d\delta_{lm}$ . The all-ones rank-one matrix has eigenvalue  $q$  on the all-ones vector and zero on its  $(q-1)$ -dimensional perpendicular complement. It follows that

$$\det(V^*V) = d^q(1+q) = d^d > 0, \quad C_q(u) := |\det \Pi^{\text{mon}}(u)|^2 = d^d \prod_{l=1}^q \left[ \frac{(d|u|)^{l/d}}{d} \Gamma(l/d) \right]^2. \quad (\text{XD18})$$

The complex determinant itself retains the oriented phase:  $\det \Pi^{\text{mon}} = \det V \prod_{l=1}^q a_l(u) e^{il\theta_0}$ .

We claim the exact global formula

$$\boxed{\det \Pi(u, c, t) = \det \Pi^{\text{mon}}(u) \exp(\mathcal{F}(c, t)/u)}. \quad (\text{XD19})$$

To prove it without assuming invertibility along the path, join  $(0, 0)$  to  $(c, t)$  by  $(sc, st)$ ,  $0 \leq s \leq 1$ . Equations (XD12) give  $\Pi' = \Pi B(s)$ , with continuous  $B(s)$ , and (XD15) gives  $\text{Tr } B(s) = \mathcal{F}'(sc, st)/u$ . For every square matrix, including a singular one,

$$\frac{d}{ds} \det \Pi = \text{Tr}(\text{adj}(\Pi) \Pi B) = \det \Pi \text{Tr } B. \quad (\text{XD20})$$

Consequently  $\det \Pi \exp(-\mathcal{F}/u)$  has derivative zero. At the starting point  $A$  is nilpotent and  $\mathcal{F}(0, 0) = 0$ , so its value is  $\det \Pi^{\text{mon}}(u)$ . This proves (XD19). Equation (XD18) proves its nonvanishing for every  $c, t$ , including every repeated-root fibre. In particular the period map

$$H_{u,t} \longrightarrow \mathbb{C}^q, \quad [P] \longmapsto \Pi \text{red}_{u,t}(P), \quad (\text{XD21})$$

is an isomorphism with inverse given by  $\Pi^{-1}$  followed by the specified degree-below- $q$  section.

#### 1.4 Exact determinant transport to the arithmetic representative

Retain the original positive Gram  $G_N$  on the cyclic coefficient space  $E$ , its determinant  $V_N = \det G_N$ , and the original maps  $r_N : E \longrightarrow \mathcal{B}^{\otimes k}$  and  $j_E$  on the admitted test-function domain with  $j_E r_N = I_E$ . Identify  $E$  with  $H_{u,t}$  by their specified ordered remainder bases. Then the actual maps

$$r_N(E) \longrightarrow \mathbb{C}^q, \quad x \longmapsto \Pi j_E x, \quad \mathbb{C}^q \longrightarrow r_N(E), \quad y \longmapsto r_N \Pi^{-1} y \quad (\text{XD22})$$

are inverse: their composites are respectively  $r_N j_E$  on  $r_N(E)$  and  $\Pi(j_E r_N) \Pi^{-1}$  on  $\mathbb{C}^q$ . On the full admitted domain the first observation has precisely  $\ker j_E$  as its kernel; all original theta boundaries killed by  $j_E$  remain in that kernel.

With the explicitly fixed coordinate metric  $I_q$  on the target, put

$$H_\Pi = \Pi^* \Pi, \quad B_N^{\text{per}} = G_N^{-1} H_\Pi. \quad (\text{XD23})$$

These operators satisfy  $G_N B_N^{\text{per}} = (B_N^{\text{per}})^* G_N = H_\Pi > 0$ . Taking determinants in (XD19) gives the exact identities

$$\det H_\Pi = C_q(u) \exp\left(2 \text{Re} \frac{\mathcal{F}(c, t)}{u}\right), \quad (\text{XD24})$$

$$\boxed{\log \det B_N^{\text{per}} = \log C_q(u) + 2 \text{Re} \frac{\mathcal{F}(c, t)}{u} - \log V_N}. \quad (\text{XD25})$$

Every determinant in (XD24)–(XD25) is positive real; these logarithms therefore require no branch choice. At the same  $u, c, t$  for two admitted degrees  $i \leq j$ , subtraction proves

$$\boxed{\log \det B_j^{\text{per}} - \log \det B_i^{\text{per}} = \log(V_i/V_j)}. \quad (\text{XD26})$$

Thus the common polynomial–exponential factor is now calculated exactly while the full original endpoint cost is retained by the displayed pair of positive comparison operators. No uniform upper estimate on  $V_i/V_j$  is inferred from this determinant identity.

## 2 SGA trace, exact constituent curvature, and the derived collision fibre

This calculation extends the constituent period map in the delivered *SGA Trace Period Control*, equations (54a)–(54c). Its period determinant is the same determinant already proved in (XD1)–(XD26), with every original coefficient, ray and phase retained. We calculate the full normal derivative, its kernel and its norm. The result applies to the actual invariant constituents, including repeated primary factors.

### 2.1 The retained period map and arithmetic source

Keep the original monic polynomial and its increasing-power frame:

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S, \quad q \geq 1, \quad e_0 = [1], \quad \ell([P]) = [S^{q-1}] \bmod_{\chi} P. \quad (\text{SC1})$$

Thus  $\ell : E \rightarrow \mathbb{C}$  is a specified linear functional and  $\mathcal{R} = e_0 \ell : E \rightarrow E$  has rank one. Fix  $u \in \mathbb{C}^*$ , one argument of  $u$ , and the actual potential and contours

$$\Phi_t(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS, \quad \theta_j = \frac{\arg u + \pi + 2\pi j}{q+1}, \quad \Gamma_j = -\ell_0^{\text{ray}} + \ell_j^{\text{ray}}. \quad (\text{SC2})$$

Here  $\ell_j^{\text{ray}}$  is the outward ray  $S = r e^{i\theta_j}$ ,  $r \geq 0$ ; it is distinct from the coefficient functional  $\ell$ . The period matrix is

$$\Pi_{jb}(t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS, \quad 1 \leq j \leq q, \quad 0 \leq b < q. \quad (\text{SC3})$$

The full convergence and determinant proof is (XD4)–(XD21). To state its precise retained inputs here, on a compact coefficient set  $K$  its tail is bounded by  $r^L \exp(-r^{q+1}/(2(q+1)|u|))$  after  $r \geq \max(1, 2(q+1)M_K)$ , where  $M_K = \sum_a \sup_K |c_a|/(a+1) + \sup_K |t|$ . Differentiation is therefore valid in every coefficient direction. Integration by parts cancels the two origin terms, and reduction of  $S^q$  uses  $[\chi - t] = 0$  with no derivative correction. It follows that

$$\Pi'(t) = -\frac{1}{u} \Pi(t)(A + t\mathcal{R}). \quad (\text{SC4})$$

The proven determinant is

$$\det \Pi(t) = \det \Pi_{\text{mon}}(u) \exp\left(\frac{\text{Tr } \Phi_t(A + t\mathcal{R})}{u}\right), \quad |\det \Pi_{\text{mon}}(u)|^2 = (2\pi|u|)^q > 0. \quad (\text{SC5})$$

The complex initial matrix is explicitly

$$(\Pi_{\text{mon}})_{j,b} = \frac{((q+1)|u|)^{(b+1)/(q+1)}}{q+1} \Gamma\left(\frac{b+1}{q+1}\right) e^{i(b+1)\theta_0} (e^{2\pi i j(b+1)/(q+1)} - 1). \quad (\text{SC6})$$

The product form and the full complex phase in (SC5)–(SC6) are reused from (XD18)–(XD19). The evaluated squared constant is the additional gamma-product evaluation in the delivered SGA note. Here is every factor in that evaluation. Set  $d = q + 1$ . The matrix  $V_{jl} = e^{2\pi i j l/d} - 1$  has  $V^* V = d(I_q + J_q)$  by the root-of-unity sum in (XD17), so  $|\det V|^2 = d^{q+1}$ . Since  $2 \sum_{l=1}^{d-1} l/d = q$ , its column factors give

$$|\det \Pi_{\text{mon}}|^2 = d^{q+1} \frac{(d|u|)^q}{d^{2q}} \left( \prod_{l=1}^{d-1} \Gamma(l/d) \right)^2. \quad (\text{SC6a})$$

The gamma multiplication formula, DLMF (5.5.6), is

$$\Gamma(dz) = (2\pi)^{(1-d)/2} d^{dz-1/2} \prod_{a=0}^{d-1} \Gamma(z + a/d). \quad (\text{SC6b})$$

At  $z = 1/d$ , its left side and the final product factor are  $\Gamma(1) = 1$ , so the remaining product equals  $(2\pi)^{q/2}d^{-1/2}$ , which is DLMF (5.5.7). In (SC6a) the complete power of  $d$  is  $(q+1) + q - 2q - 1 = 0$ . This proves  $|\det \Pi_{\text{mon}}|^2 = (2\pi|u|)^q$  without a frame change or an omitted phase. Thus every  $\Pi(t)$  is invertible, including at collisions. The full reference is <https://dlmf.nist.gov/5.5.E7>.

For every original admitted source degree  $N$ , retain

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N, \quad r_N = \mathcal{V} R_N, \quad j_E r_N = I_E. \quad (\text{SC7})$$

The source identities and primitive have their original types  $j_E D^{(k)} = A j_E$  and  $D^{(k)} r_N - r_N A = dK_N^{\text{prim}}$ . The exact parameter action on the admitted source domain is

$$D_t^{(k)} = D^{(k)} + t r_N \mathcal{R} j_E, \quad j_E D_t^{(k)} = (A + t \mathcal{R}) j_E, \quad D_t^{(k)} r_N - r_N (A + t \mathcal{R}) = dK_N^{\text{prim}}. \quad (\text{SC8})$$

Indeed insertion of  $j_E r_N = I_E$  proves each equality by multiplication; the added terms cancel in the last difference. In particular the normal map calculated below has an explicit lift to this source, with the complete original primitive and kernel of  $j_E$  retained.

## 2.2 Every invariant constituent has a nonzero top coefficient

Let  $I : F \hookrightarrow E$  be an actual injective linear map whose image is  $A$ -invariant, with  $AI = IA_F$ , and put  $p = \dim F$ . For the moment suppose  $0 < p < q$ . The image  $I(F)$  is an ideal of  $E$ : invariance under  $A$  implies invariance under every polynomial in  $A$ , and every element of  $E$  is such a polynomial applied to  $e_0$ . Its inverse image in  $\mathbb{C}[S]$  is an ideal containing  $(\chi)$ . It is generated by a unique monic divisor  $g$  of  $\chi$ ; writing  $\chi = gh$  gives  $\deg h = p$  and  $\deg g = q - p$ . The exact map is

$$\mathbb{C}[S]/(h) \longrightarrow I(F), \quad [a]_h \longmapsto [ga]_\chi. \quad (\text{SC9})$$

It is onto by the definition of  $(g)$ , and is injective because  $gh \mid ga$  in the polynomial domain implies  $h \mid a$ . Consequently  $[g], [Sg], \dots, [S^{p-1}g]$  is an ordered basis of  $I(F)$ . Its last member is monic of degree  $q - 1$ , while every earlier member has smaller degree. Hence the restricted functional  $L = \ell I : F \rightarrow \mathbb{C}$  is onto. Also  $e_0 \notin I(F)$ , since an ideal containing the unit is the whole algebra. We have proved

$$\begin{aligned} \mathcal{R}I &= e_0 L, \quad \text{rank}(\mathcal{R}I) = 1, \quad \ker(\mathcal{R}I) = \ker L, \quad \dim \ker L = p - 1, \\ \mathbb{C}e_0 \cap I(F) &= \{0\} \quad \text{inside } E. \end{aligned} \quad (\text{SC10})$$

These arguments never replace a repeated factor by its square-free part. In the basis (SC9),  $L$  is exactly the last coordinate; in the original chosen basis of  $F$  it remains the literal row  $\ell I$ .

## 2.3 The exact normal derivative at every parameter

Give the target  $\mathbb{C}^q$  the period-coordinate Hermitian metric  $I_q$  specified in the delivered note. Define

$$\begin{aligned} Y(t) &= \Pi(t)I, \quad H(t) = Y(t)^* Y(t), \quad P(t) = Y(t)H(t)^{-1}Y(t)^*, \\ N(t) &= I_q - P(t), \quad z(t) = N(t)\Pi(t)e_0. \end{aligned} \quad (\text{SC11})$$

Every  $Y(t)$  is injective, so  $H(t) > 0$ . Multiplication shows  $P(t)^2 = P(t) = P(t)^*$  and  $P(t)Y(t) = Y(t)$ ; thus  $N(t)$  is the orthogonal projection onto the complement of the displayed period subspace. The vector  $z(t)$  is nonzero for every  $t$ : otherwise  $\Pi(t)e_0$  lies in  $\Pi(t)I(F)$ , and invertibility of  $\Pi(t)$  would give  $e_0 \in I(F)$ , contradicting (SC10).

Using invariance under the original  $A$ , equation (SC4) yields

$$Y'(t) = -\frac{1}{u}Y(t)A_F - \frac{t}{u}\Pi(t)e_0L, \quad \boxed{N(t)Y'(t) = -\frac{t}{u}z(t)L.} \quad (\text{SC12})$$

This holds for every complex  $t$ , with the same fixed constituent  $F$  of the original  $A$ . We have not assumed that  $F$  is invariant under  $A + t\mathcal{R}$ . Formula (SC12) computes exactly its failure of invariance

after passing through the original periods. For  $t \neq 0$  the normal derivative has image  $\mathbb{C}z(t)$ , kernel  $\ker L$  and rank one. At  $t = 0$  it vanishes. Differentiating (SC4) once more gives

$$Y''(0) = \frac{1}{u^2}\Pi(0)A^2I - \frac{1}{u}\Pi(0)\mathcal{R}I, \quad N(0)Y''(0) = -\frac{1}{u}z(0)L. \quad (\text{SC13})$$

The first term has image inside  $Y(0)F$  because  $A^2I = IA_F^2$ . The normal acceleration therefore has rank exactly one, image  $\mathbb{C}z(0)$  and kernel  $\ker L$ , including all repeated-root cases.

The actual original source lift of its parameter defect is  $r_N e_0 L : F \rightarrow r_N(E)$ . The map  $x \mapsto N(t)\Pi(t)j_E x$  sends it to  $z(t)L$ . Thus its source, coefficient and period forms have the commuting factorization

$$F \xrightarrow{L} \mathbb{C} \xrightarrow{a \mapsto ar_N e_0} r_N(E) \xrightarrow{j_E} E \xrightarrow{N(t)\Pi(t)} \mathbb{C}^q. \quad (\text{SC14})$$

At the original Split-Zero support, every linear arrow lifts by  $x^\bullet \mapsto f(x)^\bullet$  and  $\tau \mapsto \tau$ . In particular  $\ker L$  maps to the supported zero; external absence retains its distinct image  $\tau$ . No new zero-adjunction is used.

## 2.4 Curvature is an exact rank-one expression

Write  $\partial_t = \frac{1}{2}(\partial_x - i\partial_y)$  and  $\partial_{\bar{t}} = \frac{1}{2}(\partial_x + i\partial_y)$  for  $t = x + iy$ . Since  $Y$  is holomorphic,  $\partial_t H = Y^*Y'$ ,  $\partial_{\bar{t}} H = Y'^*Y$  and  $\partial_{\bar{t}}\partial_t H = Y'^*Y'$ . Differentiating  $H^{-1}H = I_p$  gives  $\partial_{\bar{t}}H^{-1} = -H^{-1}Y'^*YH^{-1}$ . The determinant derivative and cyclic invariance of trace therefore give

$$\begin{aligned} \partial_{\bar{t}}\partial_t \log \det H &= \text{Tr}(H^{-1}Y'^*Y' - H^{-1}Y'^*YH^{-1}Y^*Y') \\ &= \text{Tr}(H^{-1}Y'^*NY'). \end{aligned} \quad (\text{SC15})$$

There is no logarithm-branch choice:  $\det H > 0$ . Insert the full normal map (SC12), use  $N^* = N = N^2$ , and retain the rank-one factors to obtain

$$\partial_{\bar{t}}\partial_t \log \det H(t) = |t|^2 c_F(t), \quad c_F(t) = \frac{\|z(t)\|_{I_q}^2}{|u|^2} L H(t)^{-1} L^* > 0. \quad (\text{SC16})$$

Strict positivity follows because  $z(t) \neq 0$ ,  $L \neq 0$  and  $H(t)^{-1} > 0$ . Thus this curvature vanishes exactly at  $t = 0$  and vanishes there to precisely quadratic order; it is strictly positive at every  $t \neq 0$ . For the dual determinant line with metric  $(\det H)^{-1}$  the convention  $c_1 = -(i/(2\pi))\partial\bar{\partial} \log h$  gives

$$c_1 = \frac{i}{2\pi} |t|^2 c_F(t) dt \wedge d\bar{t}. \quad (\text{SC17})$$

In these coordinates  $i dt \wedge d\bar{t} = 2 dx \wedge dy$ ; this records the orientation and factor explicitly.

For completeness the exact formula supplies a convergent Taylor statement. Entries of  $\Pi$  have convergent complex power series; entries of  $H$ ,  $P$ ,  $N$  and  $c_F$  are real analytic because  $\det H$  is everywhere positive. On any closed disk  $|t| \leq \varepsilon$ , continuity of the first real derivatives gives a finite bound  $M_\varepsilon = \sup \|\nabla c_F\|$ . Integrating the derivative along the real segment  $s \mapsto st$  proves  $|c_F(t) - c_F(0)| \leq M_\varepsilon |t|$ . Consequently

$$\left| \partial_{\bar{t}}\partial_t \log \det H(t) - |t|^2 c_F(0) \right| \leq M_\varepsilon |t|^3. \quad (\text{SC18})$$

This estimate follows from the actual analytic functions in (SC11), and (SC16) supplies the stronger identity at all parameters.

One may check the projector's first derivatives directly:  $\partial_t P = NY'H^{-1}Y^*$  and  $\partial_{\bar{t}} P = YH^{-1}Y'^*N$ . They vanish at zero by (SC12), explaining the retained stationary tangent space at that parameter without deleting its acceleration.

## 2.5 Both original theta and period metrics remain explicit

On  $F$  retain the positive original theta Gram  $G_{N,F} = I^* G_N I$  and the positive comparison operator

$$B_{N,F}(t) = G_{N,F}^{-1} H(t), \quad \mathcal{O}_{N,F}(t) = \log \frac{\det H(t)}{\det G_{N,F}}. \quad (\text{SC19})$$

Here  $G_{N,F}$  is fixed as  $t$  varies. Thus (SC15)–(SC18) hold unchanged with  $\mathcal{O}_{N,F}$  in place of  $\log \det H$ . For real  $t$  at zero and  $u > 0$ , (SC12) gives  $H'(0) = -(A_F^* H(0) + H(0) A_F)/u$ , hence

$$-u \mathcal{O}'_{N,F}(0) - kp = 2 \operatorname{Re} \operatorname{Tr} A_F - kp. \quad (\text{SC20})$$

The original constant  $k$  and its generalized eigenspace multiplicities are retained. On the full positive-real-defect constituent this is the existing aggregate defect  $L_{h,k}$ .

The squared Hilbert–Schmidt norm of the normal acceleration in the original domain metric, and its one nonzero squared singular value, are both

$$a_{N,F} = \frac{\|z(0)\|_{I_q}^2}{|u|^2} L G_{N,F}^{-1} L^*. \quad (\text{SC21})$$

Indeed the adjoint of  $zL : (F, G_{N,F}) \rightarrow (\mathbb{C}^q, I_q)$  is  $G_{N,F}^{-1} L^* z^*$ , so the product and its trace give (SC21). The unique nonzero eigenvalue equals its trace because the product has rank one. The curvature coefficient is instead the exact observer

$$c_F(t) = \frac{\|z(t)\|_{I_q}^2}{|u|^2} L B_{N,F}(t)^{-1} G_{N,F}^{-1} L^*. \quad (\text{SC22})$$

This follows from  $H = G_{N,F} B_{N,F}$ ; it is the required map between the two metric measurements. It does not replace either metric by an unweighted coordinate norm.

Finally transport the original metric on the whole period target by  $\tilde{G}_N = \Pi^{-*} G_N \Pi^{-1}$ . Its projection onto  $YF$  is

$$P_N^\theta = Y G_{N,F}^{-1} Y^* \tilde{G}_N = \Pi I G_{N,F}^{-1} I^* G_N \Pi^{-1}. \quad (\text{SC23})$$

Direct multiplication proves its idempotence and its  $\tilde{G}_N$ -self-adjointness. With  $z_N^\theta = (I_q - P_N^\theta) \Pi e_0$ , its squared target norm is the literal theta Schur complement

$$\|z_N^\theta\|_{\tilde{G}_N}^2 = e_0^* G_N e_0 - e_0^* G_N I G_{N,F}^{-1} I^* G_N e_0 > 0. \quad (\text{SC24})$$

The right side is positive because it is the norm of the nonzero  $G_N$ -orthogonal projection of  $e_0$  away from  $I(F)$ . For this moving target metric  $Y^* \tilde{G}_N Y = G_{N,F}$  is constant. Its differentiated metric terms are explicitly

$$\partial_t \tilde{G}_N = \Pi^{-*} G_N (A + t\mathcal{R}) \Pi^{-1} / u, \quad \partial_{\bar{t}} \tilde{G}_N = \Pi^{-*} (A + t\mathcal{R})^* G_N \Pi^{-1} / \bar{u}. \quad (\text{SC25})$$

They cancel  $Y^* \tilde{G}_N Y'$  and its conjugate derivative in the product rule. Thus the constant pulled-back theta Gram and the nonzero curvature (SC16) are connected by the exact metric congruence (SC23)–(SC25), with every term retained.

For  $F = 0$ , use the empty Gram determinant 1, empty trace 0 and the zero normal map. For  $F = E$ ,  $P = I_q$  and  $N = 0$  identically, so (SC15) is zero: the full determinant logarithm in (SC5) is the real part of a holomorphic function plus its stated constant. The strictly positive coefficient in (SC16) belongs precisely to  $0 < p < q$ . This curvature and the original theta comparison are fully calculated observables; no uniform arithmetic upper bound is asserted by these finite-dimensional identities.

## 2.6 The exact derived base change at a collision

The trace and the positive-degree diagonal groups in the delivered SGA note have a further explicit specialization map. Retain the whole one-parameter algebra over  $T = \mathbb{C}[t]$ :

$$\mathcal{E} = T[S]/(\chi(S) - t) \simeq \mathbb{C}[S], \quad t \mapsto \chi(S), \quad M = \mathcal{E}/\chi'\mathcal{E}. \quad (\text{SC26})$$

The isomorphism sends  $[P(t, S)]$  to  $P(\chi(S), S)$ ; its inverse sends  $S$  to its original class. Monic division proves that  $\mathcal{E}$  is free of rank  $q$  over  $T$ . Since  $\chi'$  is a nonzero polynomial over  $\mathbb{C}$ , multiplication by  $\chi'$  is injective on  $\mathcal{E} \simeq \mathbb{C}[S]$ . Consequently

$$0 \longrightarrow \mathcal{E} \xrightarrow{\chi'} \mathcal{E} \longrightarrow M \longrightarrow 0 \quad (\text{SC27})$$

is a free  $T$ -resolution of  $M$ . The diagonal resolution gives  $HH_0(\mathcal{E}/T) = \mathcal{E}$ ,  $HH_{2a+1}(\mathcal{E}/T) = M$  and  $HH_{2a+2}(\mathcal{E}/T) = 0$  for  $a \geq 0$ . For clarity, that resolution is free over  $\mathcal{B} = \mathcal{E} \otimes_T \mathcal{E}$  with alternating maps  $X - Y$  and  $\mathcal{C}_H = (H(X) - H(Y))/(X - Y)$ , followed by the multiplication augmentation. Over  $(T[Y]/H(Y))[X]$  both factors are monic and therefore non-zero-divisors; their product is  $H(X)$  modulo  $H(Y)$ . Cancellation proves each alternating kernel equals the preceding image. Tensoring by the diagonal sets  $X = Y$  and gives the maps  $0, \chi', 0, \chi', \dots$ , proving the stated groups.

Fix  $t_0 \in \mathbb{C}$ , put  $\kappa = T/(t - t_0)$  and  $E_{t_0} = \mathbb{C}[S]/(\chi - t_0)$ . Tensoring (SC27) gives the complete kernel-and-cokernel sequence

$$\begin{aligned} 0 \longrightarrow \text{Tor}_1^T(M, \kappa) \longrightarrow E_{t_0} \xrightarrow{\chi'} E_{t_0} \\ \longrightarrow M \otimes_T \kappa \longrightarrow 0. \end{aligned} \quad (\text{SC28})$$

It computes the actual specialized diagonal groups:

$$\begin{aligned} HH_{2a+1}(E_{t_0}/\mathbb{C}) &= M \otimes_T \kappa = E_{t_0}/(\chi'), \\ HH_{2a+2}(E_{t_0}/\mathbb{C}) &= \text{Tor}_1^T(M, \kappa) = \text{ann}_{E_{t_0}}(\chi'). \end{aligned} \quad a \geq 0. \quad (\text{SC29})$$

Every arrow is induced by the same polynomial multiplication, so these are equivariant maps for the original  $S$ -action.

There is also a representative-level description of the Tor map. Resolve  $\kappa$  by  $[T \xrightarrow{t-t_0} T]$  to identify  $\text{Tor}_1^T(M, \kappa)$  with the kernel of  $t - t_0$  on  $M$ . If  $[v]_{\chi'}$  belongs to this kernel, write

$$(\chi - t_0)v = \chi'w. \quad [v]_{\chi'} \mapsto [w]_{\chi-t_0}. \quad (\text{SC30})$$

Replacing  $v$  by  $v + \chi'a$  replaces  $w$  by  $w + (\chi - t_0)a$ , so the map is well-defined. Its image is annihilated by  $\chi'$ . If its image is zero,  $w = (\chi - t_0)a$  and cancellation of the nonzero polynomial  $\chi - t_0$  gives  $v = \chi'a$ , proving injectivity. Conversely an annihilated class  $[w]_{\chi-t_0}$  satisfies  $\chi'w = (\chi - t_0)v$  for a polynomial  $v$ , giving its preimage. These operations commute with multiplication by  $S$ .

At a root  $\lambda$  of multiplicity  $m$  in  $\chi - t_0$ , its local algebra is  $\mathbb{C}[x]/(x^m)$ ,  $x = S - \lambda$ , and  $\chi' = mx^{m-1}a(x)$  with the original invertible local factor  $a$  (this equality is in the local algebra: before reduction, the derivative of  $x^m a(x)$  also contains the term  $x^m a'(x)$ ). The odd group has local module  $\mathbb{C}[x]/(x^{m-1})$  and the even group has local module  $(x)/(x^m)$ ; both have dimension  $m - 1$ , with their full induced  $S = \lambda + x$  action. The map  $[b]_{x^{m-1}} \mapsto [xb]_{x^m}$  is an explicit module isomorphism between these local modules, using the stated local coordinate. For  $m = 1$  both groups vanish. Thus the zero even group of the family and the nonzero even group of a collision fibre are related by the exact Tor term (SC28), not by omitting specialization data. The coefficient maps in the residue construction commute with base change; this calculation retains the additional derived term required when their kernels are specialized. Lifting every arrow at its original Split-Zero support keeps its receiving supported zero and external absence distinct as in (SC14).

## 2.7 The exact neighboring-minor identity and its theta factors

The ideal basis in (SC9) supplies three actual polynomial inclusions. Write their increasing-power coefficient matrices as

$$J_g = [g, Sg, \dots, S^{p-1}g], \quad J_- = [g, Sg, \dots, S^{p-2}g], \quad J_+ = [J_g, e_0]. \quad (\text{SC31})$$

For  $p = 1$ ,  $J_-$  has zero columns. All three are injective on their stated coefficient spaces; the first two assertions follow from their distinct monic leading degrees, and the last from  $e_0 \notin I(F)$ . Let  $C = (I^*I)^{-1}I^*J_g$ . Since the columns of  $J_g$  lie in  $\text{im } I$ , this satisfies  $IC = J_g$ ; it is invertible because both matrices are bases of that same subspace. The exact frame equations are

$$H_g(t) := (\Pi J_g)^* \Pi J_g = C^* H(t) C, \quad LC = (0, \dots, 0, 1), \quad h_g(t) = |\det C|^2 \det H(t), \quad (\text{SC32})$$

where  $h_J(t) = \det((\Pi J)^* \Pi J)$  for each of the three inclusions. The scalar  $|\det C|^2$  has been retained; it is independent of  $t$ , so it has zero logarithmic derivatives.

The Gram of  $\Pi J_+$  has block form

$$\begin{pmatrix} H_g & (\Pi J_g)^* \Pi e_0 \\ (\Pi e_0)^* \Pi J_g & \|\Pi e_0\|^2 \end{pmatrix}. \quad (\text{SC33})$$

Eliminate its off-diagonal blocks using triangular matrices of determinant one. Its lower-right Schur complement is the squared norm of the orthogonal projection of  $\Pi e_0$  away from  $\text{im } \Pi J_g$ , which is  $\|z(t)\|^2$  by (SC11). Thus  $h_+ = h_g \|z\|^2$ . The last diagonal entry of  $H_g^{-1}$ , computed by the adjugate formula, is its upper-left principal minor divided by  $h_g$ , namely  $h_-/h_g$ . By (SC32) that entry is exactly  $LH^{-1}L^*$ . Formula (SC16) becomes

$$\partial_{\bar{t}} \partial_t \log h_g(t) = \frac{|t|^2}{|u|^2} \frac{h_+(t) h_-(t)}{h_g(t)^2}. \quad (\text{SC34})$$

This identity retains all three original polynomial maps. It is the precise neighboring-minor equation joining the period calculation to the source-volume and Toda calculations; it does not identify differently ordered or differently weighted families.

For each same inclusion  $J = J_-, J_g, J_+$ , define its original theta volume and its positive comparison determinant by

$$v_{N,J} = \det(J^* G_N J), \quad b_{N,J}(t) = \det((J^* G_N J)^{-1} J^* \Pi(t)^* \Pi(t) J). \quad (\text{SC35})$$

Injectivity and positivity show that both numbers are positive. For the empty inclusion both are 1. Taking the determinant of the literal product gives  $h_J = v_{N,J} b_{N,J}$ , and therefore

$$c_F(t) = \frac{1}{|u|^2} \frac{v_{N,J_+} v_{N,J_-}}{v_{N,J_g}^2} \frac{b_{N,J_+}(t) b_{N,J_-}(t)}{b_{N,J_g}(t)^2}. \quad (\text{SC36})$$

Every original theta mass and coefficient inclusion remains in the  $v$  factors; every period comparison remains in the  $b$  factors. This proves the metric dictionary required to use (SC34) within that original source programme.

For  $p = q - 1$ , the divisor  $g$  is monic linear. The square matrix  $J_+ = [g, Sg, \dots, S^{q-2}g, e_0]$  has determinant  $(-1)^{q-1}$ : expand along its last column, and the remaining minor on rows  $1, \dots, q - 1$  is upper triangular with every diagonal entry 1. Thus, with that sign retained,

$$\det(\Pi J_+) = (-1)^{q-1} \det \Pi, \quad h_+(t) = (2\pi|u|)^q \exp\left(2 \operatorname{Re} \frac{\operatorname{Tr} \Phi_t(A + t\mathcal{R})}{u}\right). \quad (\text{SC37})$$

The top minor in (SC34) is therefore the explicitly evaluated full determinant in this codimension-one case. The remaining two minors and their exact theta factors are still those in (SC31)–(SC36).

### 3 The original Laplacian control along the exact period deformation

We calculate the control parameter for the actual companion deformation, and then transport its complete numerical range through the periods. All metrics and coefficient vectors below are the original ones.

Fix the original cyclic algebra, admitted degree and positive Gram

$$\begin{aligned} E = \mathbb{C}[S]/(\chi), \quad q = \deg \chi \geq 1, \quad A = M_S, \quad G = G_N > 0, \quad k \in \mathbb{R}, \\ e_0 = [1], \quad \ell[P] = [S^{q-1}] \operatorname{rem}_\chi P. \end{aligned} \quad (\text{LC1})$$

Here  $\chi$  is the original monic polynomial,  $N \in \mathbb{Z}$  satisfies  $N \geq q - 1$ , and the deformation parameter is  $t \in \mathbb{C}$ . Put  $R = e_0 \ell$ ,  $A_t = A + tR$ , and retain the actual control form and relative, Laplacian and shifted endomorphisms

$$W_t = A_t^* G + G A_t - kG, \quad X_t = G^{-1} W_t, \quad L_t = A_t^2 - kA_t, \quad B_t = A_t - \frac{k}{2} I. \quad (\text{LC2})$$

The control form has value  $v^* W_t w = \langle v, X_t w \rangle_G$ ; thus its exact form-to-operator map is  $X_t = G^{-1} W_t$ . Here  $X_t$  is self-adjoint for  $\langle v, w \rangle_G = v^* G w$ . Its finite operator norm is defined by the supremum over the original nonzero vectors. Write  $\epsilon_N = \|X_0\|_G$ . This is an actual finite quantity, so  $-\epsilon_N G \preceq W_0 \preceq \epsilon_N G$  follows from the spectral theorem for the displayed positive metric.

For the original reflection-stable source, its complete rank-two calculation (TVB.22)–(TVB.30) evaluates this same quantity. In the unchanged source-orthogonal columns  $b_j$  and norms  $\omega_j$ , put

$$a = b_N^* G b_N, \quad d_+ = b_{N+1}^* G b_{N+1}, \quad z_N = b_N^* G b_{N+1}, \quad D_N = \det K_N.$$

For every integer  $j \geq -1$  retain  $K_j = \sum_{r=0}^j b_r b_r^* / \omega_r$ , with the empty sum  $K_{-1} = 0$  on  $E$ , and put  $D_j = \det K_j$ . The omitted-column kernel and determinant are  $\tilde{K}_N = K_{N-1} + b_{N+1} b_{N+1}^* / \omega_{N+1}$  and  $\tilde{D}_N = \det \tilde{K}_N$ . At the first admitted degree  $N = q - 1$ , the first  $q - 1$  monic remainder columns are independent, so  $K_{q-2}$  has rank  $q - 1$  and  $D_{q-2} = 0$ . No inverse of this boundary kernel is used. The retained identities give

$$\epsilon_N^2 = \frac{ad_+ - |z_N|^2}{\omega_N^2} = \alpha_{N+1} \frac{\Delta_N}{D_N}, \quad \alpha_{N+1} = \frac{\omega_{N+1}}{\omega_N}, \quad \Delta_N = D_{N+1} + D_{N-1} - D_N - \tilde{D}_N. \quad (\text{LC3})$$

The omitted-column determinant  $\tilde{D}_N$ , every boundary case and its positive source-minor sum are defined and proved in that full chapter. To check the radius here directly, the original control is  $W_0 = G(b_{N+1} b_N^* + b_N b_{N+1}^*) G / \omega_N$ . Its relative self-adjoint operator has rank at most two and trace zero;  $z_N$  is purely imaginary by the retained reflection phase. Expansion of its squared trace gives  $2(ad_+ - |z_N|^2) / \omega_N^2$ , so its two possible nonzero eigenvalues are exactly  $\epsilon_N, -\epsilon_N$ . The two-column determinant identity proves the second equality in (LC3). No estimate or limiting assumption is needed to evaluate it.

#### 3.1 The exact additional radius of the companion term

The adjoint of  $R$  for the original metric is  $R^{\dagger G} = G^{-1} \ell^* e_0^* G$ . Thus

$$X_t - X_0 = tR + \bar{t} R^{\dagger G}. \quad (\text{LC4})$$

For  $q > 1$ ,  $\ell e_0 = 0$ , hence  $R^2 = 0$ . Put

$$f = G^{-1} \ell^*, \quad a_0 = e_0^* G e_0 > 0, \quad b_0 = \ell G^{-1} \ell^* > 0, \quad \sigma_N = \sqrt{a_0 b_0}. \quad (\text{LC5})$$

The columns  $e_0, f$  are  $G$ -orthogonal because  $e_0^* G f = \bar{\ell} e_0 = 0$ . They are independent and their Gram is exactly  $\operatorname{diag}(a_0, b_0)$ . In these original unscaled columns the matrix of (LC4) is

$$\begin{pmatrix} 0 & t b_0 \\ \bar{t} a_0 & 0 \end{pmatrix}. \quad (\text{LC6})$$

Indeed  $Rf = b_0 e_0$ ,  $Re_0 = 0$ ,  $R^{\dagger G} e_0 = a_0 f$  and  $R^{\dagger G} f = 0$ . On their  $G$ -orthogonal complement both terms vanish. The characteristic polynomial on the displayed two-dimensional space is  $x^2 - |t|^2 a_0 b_0$ . Therefore the exact additional operator norm is  $|t| \sigma_N$ ; all remaining eigenvalues are zero. This retains the two original vector lengths, with no choice of unit vectors.

For  $q = 1$ , instead  $R = I_E$ , so (LC4) equals  $2 \operatorname{Re}(t) I_E$  and its exact norm is  $2 |\operatorname{Re} t|$ . Consequently the entirely specified quantity

$$E_N(t) = \epsilon_N + \begin{cases} |t| \sigma_N, & q > 1, \\ 2 |\operatorname{Re} t|, & q = 1 \end{cases} \quad \text{satisfies} \quad \|X_t\|_G \leq E_N(t), \quad -E_N(t)G \preceq W_t \preceq E_N(t)G. \quad (\text{LC7})$$

The inequality is the triangle inequality for the same operator norm. It is not an assertion that equality always holds. The two summands in (LC7) are the actual norms just computed, and the original  $\epsilon_N$  remains available through (LC3). The full Laplacian change is also exact:

$$L_t - L_0 = t(AR + RA - kR) + t^2 R^2. \quad (\text{LC8})$$

Thus the quadratic term vanishes for  $q > 1$ , while for  $q = 1$  it is the retained scalar  $t^2 I_E$ .

### 3.2 An explicit numerical-range enclosure with no missing input

Expansion of (LC2), without a commutation assumption, proves

$$L_t^* G - GL_t = A_t^* W_t - W_t A_t, \quad GL_t = W_t B_t - B_t^* G B_t - \frac{k^2}{4} G. \quad (\text{LC9})$$

For the second equality insert  $W_t = B_t^* G + G B_t$ , so the right side is  $GB_t^2 - k^2 G/4 = GL_t$ . The first equality follows by expansion; the two middle terms  $A_t^* G A_t$  cancel.

For each unchanged nonzero  $v \in E$ , define the homogeneous observations

$$z = \frac{v^* GL_t v}{v^* G v}, \quad r = \frac{\|B_t v\|_G}{\|v\|_G}, \quad a + ib = \frac{\langle v, X_t B_t v \rangle_G}{\|v\|_G^2}. \quad (\text{LC10})$$

Cauchy-Schwarz and (LC7) give  $a^2 + b^2 \leq E_N(t)^2 r^2$ . The second identity in (LC9) gives  $\operatorname{Re} z = a - r^2 - k^2/4$  and  $\operatorname{Im} z = b$ . Write  $e = E_N(t)$ . First,  $a \leq er$  implies  $\operatorname{Re} z \leq (e^2 - k^2)/4 - (r - e/2)^2$ . Second, direct multiplication gives

$$e^2 \left( \frac{e^2 - k^2}{4} - \operatorname{Re} z \right) - (\operatorname{Im} z)^2 = (a - e^2/2)^2 + e^2 r^2 - a^2 - b^2 \geq 0.$$

We have proved the actual enclosure

$$\boxed{\operatorname{Re} z \leq \frac{E_N(t)^2 - k^2}{4}, \quad (\operatorname{Im} z)^2 \leq E_N(t)^2 \left( \frac{E_N(t)^2 - k^2}{4} - \operatorname{Re} z \right)}. \quad (\text{LC11})$$

It includes every eigenvalue by taking its nonzero eigenvector; no diagonalizability is needed. It includes the case  $E_N(t) = 0$ . Every quantity in (LC11) is supplied by the original finite metric, the monic coefficient functional and the actual deformation parameter.

### 3.3 The complete period transport

For fixed  $u \neq 0$ , the full determinant proof makes the original period matrix  $\Pi(u, t)$  invertible at every parameter. Put

$$\tilde{G} = \Pi^{-*} G \Pi^{-1}, \quad \tilde{A}_t = \Pi A_t \Pi^{-1}, \quad \tilde{L}_t = \Pi L_t \Pi^{-1}, \quad \tilde{W}_t = \Pi^{-*} W_t \Pi^{-1}. \quad (\text{LC12})$$

Direct multiplication gives  $\tilde{W}_t = \tilde{A}_t^* \tilde{G} + \tilde{G} \tilde{A}_t - k \tilde{G}$  and  $\tilde{L}_t = \tilde{A}_t^2 - k \tilde{A}_t$ . For  $x = \Pi v$ , its norm and the complete numerical-range quotient are exactly those in (LC10). Its control quotient and

adjoint-defect identity are likewise transported by the literal congruences in (LC12). Thus (LC11) applies unchanged in that period realization.

The transported fixed action is  $\Pi\Pi^{-1}$ ; the displayed deformed action differs from it by  $t\Pi R\Pi^{-1}$ , whose control contribution was computed in (LC4)–(LC7). The arithmetic source realizes precisely this change through  $D_t^{(k)} = D^{(k)} + tr_N R j_E$ , with  $j_E r_N = I$  and the original boundary primitive retained, as proved in (SC7)–(SC8). These equalities supply the source, coefficient and period maps for the same Laplacian control. To retain its complete primitive, let  $\mathcal{C}_{\text{prim}}$  be the original primitive term of the tensor complex and write  $D_0 : \mathcal{C}_{\text{prim}} \rightarrow \mathcal{C}_{\text{prim}}$  for its original Euler action of cochain degree zero. Thus  $K_N^{\text{prim}} : E \rightarrow \mathcal{C}_{\text{prim}}$  and  $D^{(k)}d = dD_0$  on this term. The original boundary observation  $j_E d = 0$  makes the added finite-rank action vanish there, so  $D_t^{(k)}d = dD_0$  as well. Put  $K = K_N^{\text{prim}}$ . Using  $D_t^{(k)}r_N = r_N A_t + dK$ , expand both terms of the square:

$$\begin{aligned} (D_t^{(k)})^2 r_N &= D_t^{(k)}(r_N A_t) + D_t^{(k)}dK \\ &= (r_N A_t + dK)A_t + dD_0 K = r_N A_t^2 + d(KA_t + D_0 K). \end{aligned}$$

Subtract  $kD_t^{(k)}r_N = kr_N A_t + kdK$  with its original sign. This proves the complete quadratic source comparison

$$\boxed{((D_t^{(k)})^2 - kD_t^{(k)})r_N - r_N L_t = d(D_0 K_N^{\text{prim}} + K_N^{\text{prim}} A_t - kK_N^{\text{prim}}).} \quad (\text{LC12a})$$

The term  $K_N^{\text{prim}} A_t = K_N^{\text{prim}} A + tK_N^{\text{prim}} R$  retains the full deformation contribution. All terms inside the displayed primitive have the same target  $\mathcal{C}_{\text{prim}}$ ; the original differential, source action and boundary class remain explicit. No smallness uniform in packet size is claimed for the fully calculated  $\epsilon_N$  or  $\sigma_N$ .

### 3.4 The exact finite comparison with constituent curvature

Let  $I : F \hookrightarrow E$  be an actual injective coefficient map with  $AI = IA_F$  and  $0 < p := \dim F < q$ . In particular  $q > 1$ , so the radius  $\sigma_N$  in (LC5) applies. Retain the original orthogonal projection and its two scalar observations:

$$\begin{aligned} G_F &= I^*GI, & P_F &= IG_F^{-1}I^*G, & L &= \ell I, & f &= G^{-1}\ell^*, \\ a_{\perp} &= e_0^*G(1 - P_F)e_0, & b_F &= LG_F^{-1}L^* = \ell IG_F^{-1}I^*\ell^*. \end{aligned} \quad (\text{LC13})$$

The injection and positive  $G$  give  $G_F > 0$ . Multiplication proves  $P_F^2 = P_F$  and  $P_F^*G = GP_F$ , and its image is exactly  $I(F)$ . Consequently  $a_{\perp} = \|(1 - P_F)e_0\|_G^2$  and  $b_F = \|P_F f\|_G^2$ , the second identity following from  $I^*Gf = I^*\ell^*$ , with the same column  $f$  as in (LC5).

Both numbers are strictly positive on this actual constituent. Indeed  $A$ -invariance implies invariance under every polynomial in  $A$ , so  $I(F)$  is an ideal of  $E$ . Its inverse image in  $\mathbb{C}[S]$  is  $(g)$  for a unique monic divisor  $g$  of  $\chi$ . Writing  $\chi = gh$ , the map  $[a]_h \mapsto [ga]_{\chi}$  is an isomorphism onto  $I(F)$ : it is onto by the ideal description, and  $gh \mid ga$  implies  $h \mid a$  by cancellation in the polynomial domain. Thus  $\deg h = p$  and the complete original ideal has basis  $[g], [Sg], \dots, [S^{p-1}g]$ . The last vector is monic of degree  $q - 1$ , proving  $L \neq 0$ ; positivity of  $G_F^{-1}$  gives  $b_F > 0$ . An ideal containing  $e_0 = [1]$  is all of  $E$ . Properness therefore gives  $e_0 \notin I(F)$  and  $a_{\perp} > 0$ . Every repeated factor remains in this ideal calculation.

At the same  $u \neq 0$  and parameter  $t$ , let  $m(t)$  and  $M(t)$  be the actual smallest and largest eigenvalues of  $B(t) = G^{-1}\Pi(t)^*\Pi(t)$ . This operator is  $G$ -self-adjoint and positive because  $v^*GB(t)v = \|\Pi(t)v\|_{I_q}^2 > 0$  for each nonzero  $v$ . Its finite spectrum therefore gives  $0 < m(t) \leq M(t) < \infty$  and the quadratic-form inequalities  $m(t)G \preceq \Pi(t)^*\Pi(t) \preceq M(t)G$ . These are evaluated finite comparison quantities, with no bound uniform in the arithmetic inputs imposed on them.

Put  $Y = \Pi I$ ,  $H_F = Y^*Y$ , and  $P^{\text{per}} = YH_F^{-1}Y^*$ , the orthogonal projection in the specified period-coordinate metric  $I_q$ . Its residual is  $z = (1 - P^{\text{per}})\Pi e_0$ . The exact comparison is

$$\boxed{m(t)a_{\perp} \leq \|z(t)\|_{I_q}^2 \leq M(t)a_{\perp}, \quad \frac{b_F}{M(t)} \leq LH_F(t)^{-1}L^* \leq \frac{b_F}{m(t)}.} \quad (\text{LC14})$$

To prove the first pair, orthogonal projection gives  $\|z\|_{I_q}^2 = \min_{x \in F} \|\Pi(e_0 - Ix)\|_{I_q}^2$ . The same minimization in the original metric has value  $a_\perp$ , attained at  $x = G_F^{-1}I^*Ge_0$ . The lower quadratic-form bound holds for every  $x$ , so its minimum is at least  $m(t)a_\perp$ . Evaluating the upper bound at the displayed original minimizer gives the upper inequality. Compression by  $I$  gives  $m(t)G_F \preceq H_F \preceq M(t)G_F$ . Inversion of positive forms, followed by the unchanged row  $L$ , proves the second pair. Explicitly, for any positive matrix  $T$  and column  $w$ , completing the square proves  $w^*T^{-1}w = \max_x (2\operatorname{Re}(w^*x) - x^*Tx)$ . Applying this formula to the three positive matrices verifies the inverse order and both scalar factors without a commutation assumption.

The coefficient of the actual constituent curvature in (SC16) is  $c_F(t) = \|z(t)\|_{I_q}^2 L H_F(t)^{-1} L^* / |u|^2$ . For completeness, the period equation and  $AI = IA_F$  give  $Y' = -Y A_F / u - t \Pi e_0 L / u$ , hence  $(1 - P^{\text{per}})Y' = -tzL/u$ . Holomorphy of  $Y$  gives  $\partial_t H_F = Y^* Y'$  and  $\partial_{\bar{t}} H_F = Y'^* Y$ ; differentiating its inverse in the determinant derivative proves  $\partial_{\bar{t}} \partial_t \log \det H_F = \operatorname{Tr}(H_F^{-1} Y'^* (1 - P^{\text{per}}) Y')$ . Substitution yields the full equality and the bounds

$$\begin{aligned} \partial_{\bar{t}} \partial_t \log \det H_F(t) &= |t|^2 c_F(t), \\ \boxed{\frac{m(t)}{M(t)} \frac{a_\perp b_F}{|u|^2} \leq c_F(t) \leq \frac{M(t)}{m(t)} \frac{a_\perp b_F}{|u|^2}} & \quad (\text{LC15}) \end{aligned}$$

Every factor in (LC14) is positive, so multiplying its appropriate two lower or two upper bounds proves (LC15). In particular the coefficient is positive even at  $t = 0$ , while the actual curvature at that parameter is zero through its retained factor  $|t|^2$ .

The comparison to the Laplacian radius has an exact two-term loss. Orthogonality of  $P_F$  gives  $a_0 = \|P_F e_0\|_G^2 + a_\perp$  and  $b_0 = b_F + \|(1 - P_F)f\|_G^2$ . Multiplying with every term retained therefore proves

$$\boxed{\sigma_N^2 - a_\perp b_F = \|P_F e_0\|_G^2 b_0 + a_\perp \|(1 - P_F)G^{-1}\ell^*\|_G^2 \geq 0.} \quad (\text{LC16})$$

Combining (LC15)–(LC16) and the exact additional norm (LC4)–(LC6) gives, on this same original constituent,

$$c_F(t) \leq \frac{M(t)}{m(t)} \frac{\sigma_N^2}{|u|^2}, \quad \partial_{\bar{t}} \partial_t \log \det H_F(t) \leq \frac{M(t)}{m(t)} \frac{\|X_t - X_0\|_G^2}{|u|^2}.$$

Thus the companion control, the original constituent projections, and its period curvature have an explicit finite comparison with the complete loss and both metric comparison eigenvalues retained. The proof assigns no uniform estimate to  $M(t)/m(t)$  or  $\sigma_N$ .

## 4 Finite sampling, residue curvature and the original control form

We retain the polynomial source, remainder map, coefficient basis and arithmetic inner product from the periodized-source construction. The following calculation propagates its proved finite-source errors to the residue insertion and the original Laplacian. The quotient curvature calculation also incorporates the exact Schur-complement argument supplied by the coordinating Zeta task. Fix the original tensor degree  $k \in \mathbb{Z}_{\geq 1}$  throughout the source construction.

Let  $E = \mathbb{C}[S]/(\chi)$ , with the original monic  $\chi$  of degree  $q \geq 1$ ,  $A = M_S$ ,  $e_0 = [1]$ ,  $\ell[P] = [S^{q-1}] \text{rem}_\chi P$ , and  $R = e_0 \ell$ . For  $N \geq q - 1$ , let  $\mathcal{P}_N$  be the original polynomial coefficient space,  $J_N : \mathcal{P}_N \rightarrow E$  the full remainder map, and  $M_N > 0$  its original source Gram. Keep

$$G = G_N = (J_N M_N^{-1} J_N^*)^{-1}. \quad (\text{FC1})$$

All adjoints denoted by  $*$  are conjugate transposes in these specified coefficient frames. For a positive form  $T$ , write  $B^{\dagger T} = T^{-1} B^* T$  and  $\|B\|_T$  for its induced operator norm.

### 4.1 The actual finite sampling error

Fix the common source degree  $D \geq N$ ,  $a > 0$ , integer  $p \geq 1$ , circle length  $L > 0$  and integer cutoff  $J \geq 1$ . In the original diagonal log coordinates  $r = \log x_k$ ,  $z_i = \log x_i - \log x_k$ , retain

$$(\mathcal{U}_k F)(r, z) = e^{(kr + \sum_i z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r), \quad \mathcal{K} = L^2(\mathbb{R}^{k-1}, dz). \quad (\text{FC2})$$

The Jacobian is  $d^k x = e^{kr + \sum_i z_i} dr dz$ , proving directly that this map preserves the original norm. Its derivative satisfies  $\mathcal{U}_k D^{(k)} = (-\partial_r + k/2) \mathcal{U}_k$ . Let  $\Psi_D$  be the image of the degree- $D$  source columns. Define

$$\begin{aligned} M_{a,\pm,D} &= (e^{\pm ar} \Psi_D)^* (e^{\pm ar} \Psi_D), \quad H_{p,D} = \int_{\mathbb{R}} (1+r^2) (\partial_r^p \Psi_D)^* (\partial_r^p \Psi_D) dr, \\ \kappa_{a,D} &= \max_{c \neq 0} \frac{c^* (M_{a,+D} + M_{a,-D}) c}{c^* M_{DC} c}, \quad \lambda_{p,D} = \max_{c \neq 0} \frac{c^* H_{p,DC}}{c^* M_{DC} c}, \\ \eta_P &= \frac{\kappa_{a,D}}{e^{aL} - 1}, \quad \eta_T = \frac{\lambda_{p,D}}{2p-1} \left( \frac{L}{2\pi J} \right)^{2p-1}, \quad \delta = \eta_P + \eta_T. \end{aligned} \quad (\text{FC3})$$

These are finite observations of the original source; the weighted forms retain  $x_k^{\pm 2a}$  and  $1 + (\log x_k)^2$  in the original coordinates. For positive budgets  $\varepsilon_P, \varepsilon_T$  with  $\varepsilon_P + \varepsilon_T < 1$ , choose the literal values of (P64):

$$\begin{aligned} L &\geq a^{-1} \log(1 + \kappa_{a,D}/\varepsilon_P), \\ J &\geq \max \left\{ 1, D, \frac{L}{2\pi} \left( \frac{\lambda_{p,D}}{(2p-1)\varepsilon_T} \right)^{1/(2p-1)} \right\}, \quad J \in \mathbb{Z}. \end{aligned} \quad (\text{FC3a})$$

Substitution into (FC3) proves  $\eta_P \leq \varepsilon_P$  and  $\eta_T \leq \varepsilon_T$ , including  $\lambda_{p,D} = 0$ . Thus  $0 \leq \delta < 1$  has been obtained from actual finite quantities. No estimate uniform in  $h, k, D$  is inserted.

Use the plus Fourier transform  $\widehat{\psi}(u) = \int \psi(r) e^{iur} dr$ . Let  $M_N(L, J)$  be the full source Gram

$$M_N(L, J)_{ab} = \frac{1}{L} \sum_{|n| \leq J} \left\langle \widehat{\Psi_N e_a}(2\pi n/L), \widehat{\Psi_N e_b}(2\pi n/L) \right\rangle_{\mathcal{K}}. \quad (\text{FC4})$$

The  $n = 0$  term is included with its original factor  $L^{-1}$ . For clarity, the estimates producing (FC3) can be checked without an inverse operation. Unfolding periodization gives  $M_N(L) - M_N = \sum_{m \neq 0} \mathcal{C}_N(mL)$ . Weighted Cauchy-Schwarz gives  $|c^* \mathcal{C}_N(s) c| \leq e^{-a|s|} \sqrt{c^* M_{a,-,NC} c^* M_{a,+,NC} c}$ . Summing both lattice directions bounds its absolute value by  $c^* (M_{a,-,N} + M_{a,+,N}) c / (e^{aL} - 1)$ . Also integration by parts and  $\int (1+r^2)^{-1} dr = \pi$  give  $\|\widehat{\Psi_N c}(u)\|^2 \leq \pi |u|^{-2p} c^* H_{p,NC} c$ . Summing the two tails  $|n| > J$ , with  $\sum_{n > J} n^{-2p} \leq J^{1-2p}/(2p-1)$ , yields exactly the coefficient of  $\lambda_{p,D}$  in (FC3). Restriction from the degree- $D$  source preserves all these quadratic-form bounds. Therefore

$$(1 - \eta_P - \eta_T) M_N \preceq M_N(L, J) \preceq (1 + \eta_P) M_N. \quad (\text{FC5})$$

The asymmetry records that the omitted Fourier tail is positive. Put  $\alpha = 1 - \eta_P - \eta_T > 0$ ,  $\beta = 1 + \eta_P$ , and retain

$$H = G_N(L, J) = (J_N M_N(L, J)^{-1} J_N^*)^{-1}. \quad (\text{FC6})$$

Completing the square proves that  $v^* G v$  is the minimum of  $c^* M_N c$  over  $J_N c = v$ , and the same statement holds for  $H$ . Taking these minima in (FC5) proves

$$\alpha G \preceq H \preceq \beta G, \quad (1 - \delta)G \preceq H \preceq (1 + \delta)G. \quad (\text{FC7})$$

The coefficient minimizers are  $C_N = M_N^{-1} J_N^* G$  and  $C_N(L, J) = M_N(L, J)^{-1} J_N^* H$ . Their difference belongs to  $\chi \mathcal{P}_{N-q}$ . Fixed monic division writes its polynomial as  $\chi Q = \sum_i h(s_i) Q_i(\mathbf{s})$ . The original tensor primitive replaces  $F_h$  by  $\phi_*$  in slot  $i$ , applies  $Q_i(D_1, \dots, D_k)$ , and has coefficient  $(-1)^{i-1}$ . The tensor differential contributes the same sign. Thus this change of representative has an explicit primitive with every original summand retained.

## 4.2 A dimension-free relative estimate for the residue coefficient

Let  $I : F \hookrightarrow E$  be an actual nonzero proper  $A$ -invariant constituent and set  $L_F = \ell I$ . For every positive form  $T$  define

$$\begin{aligned} T_F &= I^* T I, & P_{T,F} &= I T_F^{-1} I^* T, \\ a_F(T) &= \min_{x \in F} (e_0 - Ix)^* T (e_0 - Ix) = e_0^* T (1 - P_{T,F}) e_0, \\ b_F(T) &= L_F T_F^{-1} L_F^*, & \mathbf{c}_F(T) &= a_F(T) b_F(T), & \sigma(T)^2 &= (e_0^* T e_0) (\ell T^{-1} \ell^*). \end{aligned} \quad (\text{FC8})$$

Here  $q > 1$ . To verify strict positivity,  $I(F)$  is an ideal of  $E$  because it is invariant under every polynomial in  $A$ . Its polynomial preimage is  $(g)$  with  $\chi = gh$ , and the original ideal basis is  $[g], \dots, [S^{p_F-1}g]$ , where  $p_F = \deg h = \dim F$ . Its last member is monic of degree  $q - 1$ , so  $L_F \neq 0$ ; properness gives  $e_0 \notin I(F)$ . These statements prove  $a_F(T), b_F(T) > 0$  without discarding repeated factors.

The same constrained-minimum argument as in (FC7) gives  $\alpha a_F(G) \leq a_F(H) \leq \beta a_F(G)$ . Compression and inverse order give  $\beta^{-1} b_F(G) \leq b_F(H) \leq \alpha^{-1} b_F(G)$ . Inverse order follows, for example, from the exact identity  $w^* T^{-1} w = \max_x (2 \operatorname{Re} w^* x - x^* T x)$ . Multiplication proves the full coefficient estimate

$$\boxed{\frac{\alpha}{\beta} \leq \frac{\mathbf{c}_F(H)}{\mathbf{c}_F(G)} \leq \frac{\beta}{\alpha}, \quad \left| \log \frac{\mathbf{c}_F(H)}{\mathbf{c}_F(G)} \right| \leq \log \frac{\beta}{\alpha}.} \quad (\text{FC9})$$

Applying the same two form bounds to  $e_0$  and the inverse form to  $\ell^*$  also proves

$$\frac{\alpha}{\beta} \sigma(G)^2 \leq \sigma(H)^2 \leq \frac{\beta}{\alpha} \sigma(G)^2. \quad (\text{FC10})$$

Every number is evaluated in its stated original or sampled metric. Neither (FC9) nor (FC10) has a dimension factor.

For the actual period matrix  $\Pi(u, t)$ ,  $u \neq 0$ , put  $T_{\text{per}}(t) = \Pi(t)^* \Pi(t)$  and let  $m_H(t), M_H(t)$  be the least and greatest eigenvalues of  $H^{-1} T_{\text{per}}(t)$ . Their positivity follows from the nonzero full period determinant. Applying the same calculation, now between  $H$  and  $T_{\text{per}}$ , gives

$$\frac{m_H(t)}{M_H(t)} \mathbf{c}_F(H) \leq \mathbf{c}_F(T_{\text{per}}(t)) \leq \frac{M_H(t)}{m_H(t)} \mathbf{c}_F(H). \quad (\text{FC11})$$

The period equation  $\Pi' = -\Pi(A + tR)/u$  implies, with  $Y = \Pi I$  and  $P = Y(Y^* Y)^{-1} Y^*$ ,  $(1 - P)Y' = -t(1 - P)\Pi e_0 L_F/u$ . Differentiating  $\log \det(Y^* Y)$  gives  $\partial_{\bar{t}} \partial_t \log \det(Y^* Y) = \operatorname{Tr}((Y^* Y)^{-1} Y'^* (1 - P)Y')$ . Consequently the precise connection to (FC9) is

$$\partial_{\bar{t}} \partial_t \log \det(Y^* Y) = \frac{|t|^2}{|u|^2} \mathbf{c}_F(T_{\text{per}}(t)). \quad (\text{FC12})$$

Equations (FC9)–(FC12) retain the actual comparison spectrum; they do not substitute the sampled coefficient for the period coefficient.

There is also a useful exact endpoint bound. For two admitted degrees  $i \leq j$ , the original representative spaces increase, so  $G_i \succeq G_j > 0$ . Let  $\theta$  and  $\zeta_+$  be the least and greatest eigenvalues of the actual relative operator  $G_i^{-1}G_j$ . Then  $\theta G_i \preceq G_j \preceq \zeta_+ G_i$  with  $0 < \theta \leq \zeta_+ \leq 1$ . Applying the proved minimum and inverse comparisons with these two factors gives the first inequality below. All eigenvalues lie in  $(0, 1]$ , so their product is at most their least member. This proves the remaining inequalities and retains the sharper endpoint coefficient from (RCX21):

$$\left| \log \frac{\mathfrak{c}_F(G_j)}{\mathfrak{c}_F(G_i)} \right| \leq \log \frac{\zeta_+}{\theta} \leq -\log \theta \leq \log \frac{\det G_i}{\det G_j}. \quad (\text{FC13})$$

The last inequality retains the correct direction. It does not infer an upper volume estimate from a scalar curvature observation. For four original endpoints, (FC7) similarly gives the fully finite bound  $|\mathcal{B}(L, J) - \mathcal{B}| \leq 2q \log(\beta/\alpha)$ .

### 4.3 Transporting the control form through the sampling error

The control also depends on the unchanged operator, so we retain its complete error term. For  $t \in \mathbb{C}$ , retaining that tensor degree  $k$ , set

$$\begin{aligned} A_t &= A + tR, \quad T_t = A_t - \frac{k}{2}I, \quad W_G(t) = A_t^*G + GA_t - kG, \quad X_G(t) = G^{-1}W_G(t), \\ W_H(t) &= A_t^*H + HA_t - kH, \quad X_H(t) = H^{-1}W_H(t), \quad B = G^{-1}(H - G). \end{aligned} \quad (\text{FC14})$$

The matrix identities in this subsection and in (FC24)–(FC29) also hold for any real scalar  $k$  when considered on the displayed finite coefficient space; the source realization retains its given integer tensor degree. Then  $B^{\dagger G} = B$ ,  $\|B\|_G \leq \delta$ , and  $H = G(I + B)$ . No coordinate vector or inner product has been rescaled. Direct multiplication proves the exact identity

$$(I + B)X_H(t) = X_G(t) + T_t^{\dagger G}B + BT_t. \quad (\text{FC15})$$

Let  $\epsilon_G(t) = \|X_G(t)\|_G$ ,  $\epsilon_H(t) = \|X_H(t)\|_H$ , and  $\tau_G(t) = \|T_t\|_G$ . The two  $X$  operators are self-adjoint in their respective metrics, so their norms are the suprema of their absolute quadratic forms divided by the respective squared norms. For every original  $v$ ,

$$|v^*(W_H(t) - W_G(t))v| = |\langle T_tv, Bv \rangle_G + \langle Bv, T_tv \rangle_G| \leq 2\delta\tau_G(t) v^*Gv. \quad (\text{FC16})$$

Using (FC7) in both directions proves

$$\boxed{\epsilon_H(t) \leq \frac{\epsilon_G(t) + 2\delta\tau_G(t)}{\alpha}, \quad \epsilon_G(t) \leq \beta\epsilon_H(t) + 2\delta\tau_G(t).} \quad (\text{FC17})$$

Thus the control transfer retains the actual generator size as well as the relative metric error. Both quantities are finite observations of the original matrices. The exact LC additional radii  $|t|\sigma(G)$  for  $q > 1$ , and  $2|\operatorname{Re} t|$  for  $q = 1$ , remain available through (FC10).

For completeness, define the evaluated upper bound  $e = \beta\epsilon_H(t) + 2\delta\tau_G(t)$  and  $L_t = A_t^2 - kA_t$ . Expansion gives  $GL_t = W_G(t)T_t - T_t^*GT_t - k^2G/4$ . For each nonzero original vector  $v$ , put  $z = v^*GL_tv/(v^*Gv)$ ,  $r = \|T_tv\|_G/\|v\|_G$  and  $a + ib = \langle v, X_G(t)T_tv \rangle_G/(v^*Gv)$ . Equation (FC17) gives  $a^2 + b^2 \leq e^2r^2$ , while the expansion gives  $\operatorname{Re} z = a - r^2 - k^2/4$  and  $\operatorname{Im} z = b$ . The identities  $er - r^2 = e^2/4 - (r - e/2)^2$  and

$$e^2 \left( \frac{e^2 - k^2}{4} - \operatorname{Re} z \right) - b^2 = (a - e^2/2)^2 + e^2r^2 - a^2 - b^2$$

prove the actual original-metric enclosure

$$\operatorname{Re} z \leq \frac{e^2 - k^2}{4}, \quad (\operatorname{Im} z)^2 \leq e^2 \left( \frac{e^2 - k^2}{4} - \operatorname{Re} z \right). \quad (\text{FC18})$$

No division by  $e$  occurs, so the zero-radius case is included.

#### 4.4 The finite Fourier boundary and the quotient compensation

Let  $Q_J$  be the orthogonal Fourier projection to  $|n| \leq J$ , with the full  $\mathcal{K}$  fibre at each retained mode. It commutes with  $D_L = -\partial_r + k/2$ . Use the actual least-norm representative for  $M_N(L, J)$  and write

$$\begin{aligned} r^{L,J} &= \mathcal{V}_{h,k} C_N(L, J), & \mathcal{R} &= Q_J \mathcal{P}_L \mathcal{U}_k r^{L,J}, \\ \mathcal{B} &= Q_J \mathcal{P}_L \mathcal{U}_k (D^{(k)} r^{L,J} - r^{L,J} A). \end{aligned} \quad (\text{FC19})$$

The explicit polynomial-division primitive following (FC7) applies to this relation as well. Direct intertwining gives  $D_L \mathcal{R} - \mathcal{R} A = \mathcal{B}$  and  $\mathcal{R}^* \mathcal{R} = H$ . Since  $D_L^* + D_L = k$  on these smooth finite-mode columns, multiplication and then one further application of  $D_L$  prove

$$\begin{aligned} W_H(0) &= -(\mathcal{R}^* \mathcal{B} + \mathcal{B}^* \mathcal{R}), \\ (D_L^2 - k D_L) \mathcal{R} - \mathcal{R} (A^2 - k A) &= (D_L - k) \mathcal{B} + \mathcal{B} A. \end{aligned} \quad (\text{FC20})$$

In particular (FC17) transports an actual boundary control computed from (FC20) to the original Gram. Finite mode count does not assign interval enclosures to its arithmetic coefficients; such enclosures have to be proved for any numerical application.

Finally retain the exact coefficient quotient  $E/I(F)$  and a constant lift matrix  $J_F$  of its fixed basis. Then  $C = [I, J_F]$  is invertible. For  $T_{\text{per}} = \Pi^* \Pi$  the least-period-norm lift and its Gram are

$$\begin{aligned} Q(t) &= J_F - I(I^* T_{\text{per}} I)^{-1} I^* T_{\text{per}} J_F, \\ K(t) &= J_F^* T_{\text{per}} J_F - J_F^* T_{\text{per}} I(I^* T_{\text{per}} I)^{-1} I^* T_{\text{per}} J_F. \end{aligned} \quad (\text{FC21})$$

Completing the square in  $\|\Pi(J_F v + Ix)\|^2$  proves uniqueness, positivity,  $\pi Q = \pi J_F$  and  $I^* T_{\text{per}} Q = 0$ . The change  $J_F \mapsto J_F + I B_0$  cancels exactly in  $Q$  and  $K$ ; a quotient basis change  $J_F \mapsto J_F D_0$  gives  $K \mapsto D_0^* K D_0$ . Block elimination in  $C^* T_{\text{per}} C$  gives

$$\det(I^* T_{\text{per}} I) \det K = |\det C|^2 |\det \Pi|^2. \quad (\text{FC22})$$

The complete period determinant is a nonzero constant times a holomorphic exponential in  $t$  at fixed  $u$ . Its retained squared constant is  $(2\pi|u|)^q$ ; hence its log modulus squared has zero mixed curvature. Differentiating (FC22) and using (FC12) proves

$$\partial_{\bar{t}} \partial_t \log \det K(t) = -\frac{|t|^2}{|u|^2} \mathbf{c}_F(T_{\text{per}}(t)). \quad (\text{FC23})$$

Both actual curvatures vanish at  $t = 0$ , and their nonzero leading coefficients have opposite signs. More precisely applying  $\partial_{\bar{t}} \partial_t$  to (FC12) at zero gives  $\partial_{\bar{t}}^2 \partial_t^2 \log \det(I^* T_{\text{per}} I)|_0 = \mathbf{c}_F(T_{\text{per}}(0))/|u|^2$ ; the same derivative of  $\log \det K$  is its negative. The respective  $t^2 \bar{t}^2$  Taylor coefficients are these numbers divided by  $2!2!$ .

Apply the same Schur formula to the unchanged  $G$  and sampled  $H$  to obtain  $K_G, K_H$ . The fibre minima in the same quotient give  $\alpha K_G \preceq K_H \preceq \beta K_G$ . The literal comparison endomorphism  $K_G^{-1} K(t)$  satisfies  $K_G(K_G^{-1} K(t)) = K(t)$  and has log determinant curvature (FC23). This supplies the metric morphism on the original quotient, together with its finite sampling error and its exact curvature compensation. No coefficient class, generalized eigenspace or support label is removed by these identities.

#### 4.5 The exact commutator allowance for the same two metrics

The coordinating Zeta task supplied the following sharper transfer, which we prove in full for the actual  $G, H, A_t$  already defined. It retains the specific metric discrepancy that interacts with the generator. All adjoints in this subsection are for  $G$ . Put  $\mathcal{C} = G^{-1} H = I + B$ , using the same  $B$  as in (FC14). This is a positive  $G$ -self-adjoint endomorphism with spectrum in  $[\alpha, \beta]$ . This symbol denotes

the metric endomorphism; the constant quotient frame  $C = [I, J_F]$  in (FC22) remains unchanged. Subtract  $\mathcal{C}X_G(t)$  from (FC15), in the displayed order:

$$\boxed{X_H(t) - X_G(t) = \mathcal{C}^{-1}[T_t^{\dagger G}, B].} \quad (\text{FC24})$$

Indeed  $X_G = T_t^{\dagger G} + T_t$ , so the numerator of this difference is  $T_t^{\dagger G}B - BT_t^{\dagger G}$ .

Let  $\mathcal{S} = \mathcal{C}^{1/2}$  be the unique positive  $G$ -self-adjoint square root. The finite spectral theorem constructs it by replacing each positive eigenvalue of the actual  $\mathcal{C}$  by its positive square root. Uniqueness follows because a positive square root commutes with its square and, on each eigenspace of that square, positivity forces its eigenvalues to be that positive root. Thus no additional metric is selected. The exact map  $\mathcal{S} : (E, H) \rightarrow (E, G)$  is an onto isometry, since  $\mathcal{S}^*G\mathcal{S} = G\mathcal{S}^2 = G\mathcal{C} = H$ . Its inverse is the actual  $\mathcal{S}^{-1}$ . In this comparison set

$$\begin{aligned} Y_H &= \mathcal{S}X_H\mathcal{S}^{-1} = \mathcal{S}^{-1}T_t^{\dagger G}\mathcal{S} + \mathcal{S}T_t\mathcal{S}^{-1}, \\ Z &= [\mathcal{S}, T_t]\mathcal{S}^{-1}, \quad Y_H - X_G = Z + Z^{\dagger G}. \end{aligned} \quad (\text{FC25})$$

The adjoint of  $Z$  is  $\mathcal{S}^{-1}[T_t^{\dagger G}, \mathcal{S}]$ ; inserting it proves the last identity with both ordered terms present. The isometry gives  $\|Y_H\|_G = \epsilon_H(t)$ , so the reverse triangle inequality gives

$$|\epsilon_H(t) - \epsilon_G(t)| \leq \|Z + Z^{\dagger G}\|_G \leq 2\|[\mathcal{S}, T_t]\mathcal{S}^{-1}\|_G. \quad (\text{FC26})$$

The commutator  $Y = [\mathcal{S}, T_t]$  satisfies the Sylvester equation  $\mathcal{S}Y + Y\mathcal{S} = [\mathcal{C}, T_t] = [B, T_t]$ . Its exact solution is

$$[\mathcal{S}, T_t] = \int_0^\infty e^{-r\mathcal{S}}[B, T_t]e^{-r\mathcal{S}}dr. \quad (\text{FC27})$$

To prove convergence, the original  $G$ -operator norm satisfies  $\|e^{-r\mathcal{S}}\|_G \leq e^{-r\sqrt{\alpha}}$ . Differentiating the integrand and integrating its two terms gives  $\mathcal{S}Y + Y\mathcal{S} = [B, T_t]$ , with value  $[B, T_t]$  at zero and limit zero at infinity. For uniqueness, in the spectral decomposition of  $\mathcal{S}$ , each matrix block of a homogeneous solution is multiplied by the positive sum of its two eigenvalues; all such blocks must vanish. Therefore the integral equals the stated commutator, rather than a separate solution chosen for a bound. Its norm is at most  $\|[B, T_t]\|_G/(2\sqrt{\alpha})$ , while  $\|\mathcal{S}^{-1}\|_G \leq \alpha^{-1/2}$ . Substitution proves

$$\begin{aligned} \boxed{|\epsilon_H(t) - \epsilon_G(t)| \leq \|Z + Z^{\dagger G}\|_G \leq \alpha^{-1}\|[B, T_t]\|_G,} \\ [B, T_t] = [B, A] + t[B, R], \quad [B, R]v = (Be_0)\ell(v) - e_0\ell(Bv). \end{aligned} \quad (\text{FC28})$$

The scalar  $k/2$  commutes exactly, and both residue terms remain. In particular the fully evaluated number

$$e_{\text{comm}}(t) = \epsilon_H(t) + \|Z + Z^{\dagger G}\|_G \leq \epsilon_H(t) + \alpha^{-1}\|[B, A] + t[B, R]\|_G \quad (\text{FC29})$$

is another valid original-metric control radius in (FC18). Taking the lesser of this number and the radius there remains valid because both have just been proved to bound  $\epsilon_G(t)$ . If  $[B, T_t] = 0$ , its  $G$ -adjoint gives  $[T_t^{\dagger G}, B] = 0$ , and uniqueness in (FC27) gives  $[\mathcal{S}, T_t] = 0$ . Equations (FC24)–(FC26) then prove  $X_H = X_G$ ,  $Y_H = X_G$  and  $\epsilon_H = \epsilon_G$  exactly.

These identities are covariant under every invertible coordinate map  $U : E' \rightarrow E$ . Indeed  $G' = U^*GU$ ,  $H' = U^*HU$ ,  $T'_t = U^{-1}T_tU$  and  $B' = U^{-1}BU$  imply  $\mathcal{S}' = U^{-1}\mathcal{S}U$  by positive-square-root uniqueness in  $G'$ . Every commutator and integrand above transforms by this same similarity, and every norm is preserved by the stated coordinate map. Thus (FC28) is a comparison of the two original source observations with their full generator; its arithmetic commutator still requires evaluation for any numerical or uniform endpoint application.